§1. Binomial Asset Pricing Model

Problem 1.1.
Assume in the one-period binomial market that both H and T have positive probability of occurring. Show that the condition that $0 < d < 1 + r < u$ precludes arbitrage. In other words, show that if $X_0 = 0$ and

$$X_1 = \Delta_0 S_1 + (1 + r)(X_0 - \Delta_0 S_0)$$

then we cannot have X_1 strictly positive with positive probability unless X_1 is strictly negative with positive probability as well, and this is the case regardless of the choice of the number Δ_0.

Observe that

$$X_1(H) = (u - 1 - r)\Delta_0 S_0 + (1 + r)X_0 = (u - 1 - r)\Delta_0 S_0$$

and likewise $X_1(T) = (d - 1 - r)\Delta_0 S_0$. Suppose towards contradiction that $X_1(H)$ is positive and $X_1(T)$ is non-negative. Then, if $\Delta_0 > 0$, it must be that $\Delta_0 S_0 > 0$ and thus $d \geq 1 - r$, violating the condition. $\Delta_0 \neq 0$ because then $X_1(H) = 0 = X_1(T)$. If $\Delta_0 < 0$, then it must be that $u - 1 - r < 0$, also violating the condition. □

Problem 1.2.
In the one-period binomial model of Section 1.1, suppose we want to determine the price at time zero of the derivative security $V_1 = S_1$ (i.e., the derivative security pays off the stock price.) (This can be regarded as a European call with strike price K = 0). What is the time-zero price V_0 given by the risk-neutral pricing formula (1.1.10)?

Observe that

$$V_0 = \frac{1}{1 + r}[\tilde{p}V_1(H) + \tilde{q}V_1(T)] = \frac{1}{1 + r}[\tilde{p}S_1(T) + \tilde{q}S_1(H)] = S_0$$

as $\tilde{p}$ and $\tilde{q}$ are the solutions to

$$S_0 = \frac{1}{1 + r}[\tilde{p}S_1(T) + \tilde{q}S_1(H)].$$

§2. Probability Theory

Problem 2.1.
Using Definition 2.1.1, show the following.

1. If A is an event and A^c denotes its complement, then $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
2. If $A_1, A_2, ..., A_N$ is a finite set of events, then

$$\mathbb{P}\left(\bigcup_{n=1}^{N} A_n\right) \leq \sum_{n=1}^{N} \mathbb{P}(A_n)$$

If the events $A_1, A_2, ..., A_N$ are disjoint, then equality holds.

1. **Solution**: Because $\Omega = A \cup A^c$, we have $\mathbb{P}(A \cup A^c) = \mathbb{P}(\Omega) = 1$. Because $A \cup A^c = \varnothing$, $\mathbb{P}(A \cup A^c) = \mathbb{P}(A) + \mathbb{P}(A^c)$. Therefore, $\mathbb{P}(A^c) = \mathbb{P}(A \cup A^c) - \mathbb{P}(A) = 1 - \mathbb{P}(A)$. □
2. **Solution**: Define $B_1 = A_1$, $B_n = A_2 \setminus \left(\bigcup_{m=1}^{n-1} A_m\right)$. Observe that $\bigcup_{n=1}^{N} A_n = \bigcup_{n=1}^{N} B_n$ and $\mathbb{P}(B_n) \leq \mathbb{P}(A_n)$ as $B_n \subseteq A_n$. Therefore,

$$\mathbb{P}\left(\bigcup_{n=1}^{N} A_n\right) = \mathbb{P}\left(\bigcup_{n=1}^{N} B_n\right) = \sum_{n=1}^{N} \mathbb{P}(B_n) \leq \sum_{n=1}^{N} \mathbb{P}(A_n). \quad \square$$

Problem 2.2.
Consider the stock price S_3 in Figure 2.3.1.

(i) What is the distribution of S_3 under the risk-neutral probabilities $\tilde{p} = \frac{1}{2}$, $\tilde{q} = \frac{1}{2}$.

(ii) Compute $\tilde{\mathbb{E}}S_1$, $\tilde{\mathbb{E}}S_2$, and $\tilde{\mathbb{E}}S_3$. What is the average rate of growth of the stock price under $\tilde{\mathbb{P}}$?

(iii) Answer (i) and (ii) again under the actual probabilities $p = \frac{2}{3}$, $q = \frac{1}{3}$.

Problem 2.3.
Exercise 2.8 in the text!